

ECON712: Final Exam

April 16th, 2026

Question I: Heterogeneity and Causal Inference

Suppose that, for a sample of individuals, average cancer rates of smokers and non-smokers are given as follows:

Table 1: Cancer Diagnosis by Age Group and Smoking Status

Age Group (x)	Smoker	Non-Smoker
60-70	75 (80)	25 (40)
50-60	31 (40)	10 (20)
40-60	6 (10)	10 (90)
30-40	2 (5)	3 (45)

Notes: Sample size are given in brackets

Suppose that you want to estimate the following regression:

$$y_i = \beta_0 + \beta_1 D_i + \epsilon$$

where:

$$D_i = \begin{cases} 1 & \text{if person } i \text{ is a smoker} \\ 0 & \text{if person } i \text{ is a non-smoker} \end{cases} \quad y_i = \begin{cases} 1 & \text{if person } i \text{ has cancer} \\ 0 & \text{if person } i \text{ does not have cancer} \end{cases}$$

Using the information in this table, answer the following questions:

- (a) (7 points) Suppose that you use the sample to estimate the regression, calculate $\hat{\beta}_1^{OLS/MM}$. Is this an unbiased estimator of ATE? Is this a consistent estimator of ATE? Explain.
- (b) (7 points) Assume that $y(0), y(1) \perp D|x$, calculate ATE and ATT

Question II: OLS Finite Sample Properties

Assume that the model $y = X\beta + u$ satisfies the Gauss-Markov assumptions (i.e. classical assumptions), where X is a $n \times k$ matrix of observations. Define a new matrix $Z = aX$, where $a > 1$ is a scalar. Suppose a researcher draws a random sample n and obtains the following estimator:

$$\tilde{\beta} = (Z'X)^{-1}Z'y$$

Using this information answer the following question:

- (a) (5 points) Prove whether $\tilde{\beta}$ is a biased or unbiased estimator of β ?

- (b) (5 points) Calculate the variance-covariance matrix of the estimator. Which estimator do you prefer, $\tilde{\beta}$ or $\hat{\beta}$, where $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is the method-of-moment estimator derived in class? Explain.
- (c) (6 points) Suppose now that you use the OLS estimator, $\hat{\beta}$, but that there is heteroscedasticity. In particular, $E[\mathbf{u}\mathbf{u}'] = \mathbf{\Omega}$, where:

$$\mathbf{\Omega} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_N^2 \end{bmatrix}$$

Carefully explain how you would estimate the variance-covariance matrix of the OLS estimator when $\mathbf{\Omega}$ is not known (make sure to discuss the importance of the sandwich estimator in your answer). Is OLS the most efficient estimator in this case?

Question III: Panel Data and Difference-in-Differences

Consider a balanced panel dataset with n units observed over $T = 2$ periods ($t \in \{1, 2\}$). The true causal population regression is:

$$y_{it} = \beta x_{it} + \alpha_i + \lambda_t + \varepsilon_{it}, \quad i = 1, \dots, n, \quad t \in \{1, 2\},$$

where x_{it} is a scalar regressor, α_i is a **unit fixed effect** that may be correlated with x_{it} , λ_t is a **common time effect** shared by all units (e.g. a macroeconomic shock), and ε_{it} is an idiosyncratic error. Assume strict exogeneity throughout: $\mathbb{E}[\varepsilon_{it} \mid x_{i1}, x_{i2}, \alpha_i] = 0$ for $t \in \{1, 2\}$. In addition, note that since λ_t is a fixed parameter, the conditional mean of the outcome net of the unit effect and the regressor is period-specific:

$$\mathbb{E}[y_{it} - \beta x_{it} - \alpha_i \mid \mathbf{x}_i, t = 1] = \lambda_1, \quad \mathbb{E}[y_{it} - \beta x_{it} - \alpha_i \mid \mathbf{x}_i, t = 2] = \lambda_2,$$

where λ_1 and λ_2 are not restricted to be equal, capturing a common time effect that may differ across periods.

Stack all $2n$ observations into the $(2n \times 1)$ vectors $\mathbf{y}$, $\mathbf{x}$, $\boldsymbol{\varepsilon}$, and $\boldsymbol{\lambda}$ (where λ_t is repeated n times for each period), and let $\mathbf{D}$ be the $(2n \times n)$ matrix of unit dummies. Define the composite error $\mathbf{u} \equiv \mathbf{D}\boldsymbol{\alpha} + \boldsymbol{\lambda} + \boldsymbol{\varepsilon}$. The true model in stacked form is:

$$\mathbf{y} = \mathbf{x}\beta + \mathbf{D}\boldsymbol{\alpha} + \boldsymbol{\lambda} + \boldsymbol{\varepsilon}.$$

- (a) (7 points) A researcher who ignores both α_i and λ_t runs pooled OLS of $\mathbf{y}$ on $\mathbf{x}$ and a constant term, treating $\mathbf{u}$ as the error term. Derive the pooled OLS estimator $\hat{\beta}^{OLS}$ in matrix form as a function of the true population parameter β and fixed effect terms. Carefully discuss why the pooled OLS might be inconsistent (**you do not need to show the math**).

- (b) (7 points) The researcher now attempts to correct for unit fixed effects by applying the within transformation. For each unit i , define demeaned variables:

$$\tilde{y}_{it} = y_{it} - \bar{y}_i, \quad \tilde{x}_{it} = x_{it} - \bar{x}_i,$$

where $\bar{y}_i = \frac{1}{2}(y_{i1} + y_{i2})$ and $\bar{x}_i = \frac{1}{2}(x_{i1} + x_{i2})$. By the Frisch–Waugh–Lovell theorem, running OLS on the demeaned variables $\tilde{\mathbf{y}}$ and $\tilde{\mathbf{x}}$ is equivalent to partialling out unit fixed effects from both $\mathbf{y}$ and $\mathbf{x}$. Write the demeaned model for $\tilde{y}_{it}$ in terms of $\tilde{x}_{it}$, $\tilde{\varepsilon}_{it}$, and the demeaned time effects $\tilde{\lambda}_t \equiv \lambda_t - \bar{\lambda}$, where $\bar{\lambda} = \frac{1}{2}(\lambda_1 + \lambda_2)$. Show that the common time effect does *not* vanish after the within transformation and express $\tilde{\lambda}_1$ and $\tilde{\lambda}_2$ in terms of $\delta \equiv \lambda_2 - \lambda_1$. Provide an intuitive explanation of why controlling for unit fixed effects might not be sufficient.

- (c) (7 points) Suppose now that individuals i belong to one of two groups $s \in \{0, 1\}$, a control group ($s = 0$) and a treatment group ($s = 1$), observed over $t \in \{1, 2\}$. Let D_s be a dummy variable equal to one if the individual belongs to the treatment group and zero otherwise, and let λ_t be a dummy variable equal to one if the observation is in the second period and zero otherwise. Consider the following population regression:

$$y_{ist} = \alpha + \beta_1 D_s + \beta_2 \lambda_t + \beta_3 (D_s \cdot \lambda_t) + u_{ist}.$$

Discuss the difference-in-differences estimator of β_3 , the assumptions required for β_3 to have a causal interpretation, and express $\hat{\beta}_3$ in terms of group-period sample means $\bar{y}_{s,t}$.

Question IV: Linear transformation of variables (8 points)

Assume the regression model:

$$\mathbf{y} = \boldsymbol{\iota}_n \hat{\gamma} + \mathbf{X} \hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}$$

where

- $\mathbf{y}$ is a $nx1$ vector of dependent variables
- $\boldsymbol{\iota}_n$ is an $nx1$ vector of ones,
- $\mathbf{X}$ is a nxk matrix of regressors,
- $\hat{\gamma}$ is a scalar OLS/MM estimate
- $\hat{\boldsymbol{\beta}}$ is a $kx1$ vector of OLS/MM estimates
- $\hat{\mathbf{u}}$ is a $nx1$ vector of residuals

Define the following transformed variables:

$$\mathbf{z} = \mathbf{y} - \boldsymbol{\iota}_n \bar{y}$$

and

$$\mathbf{W} = \mathbf{X} - \boldsymbol{\iota}_n \bar{\mathbf{x}}^T$$

with

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad , \quad \bar{\mathbf{x}} = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_k \end{pmatrix} \quad , \quad \bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$$

Suppose that a researcher uses OLS/MM to estimate the following regression instead:

$$\mathbf{z} = \mathbf{W} \hat{\boldsymbol{\alpha}} + \hat{\boldsymbol{\epsilon}}$$

Derive the relationship between $\hat{\boldsymbol{\alpha}}$ and $\hat{\boldsymbol{\beta}}$. Are the residuals from these two regressions equal?

Question V: Local Average Treatment Effect

Consider a setting with a binary outcome Y_i , a binary treatment $D_i \in \{0, 1\}$, and a binary instrument $Z_i \in \{0, 1\}$. Let $Y_i(d)$ denote the potential outcome under treatment status d , and let $D_i(z)$ denote the treatment status individual i would take under instrument value z . The Wald estimator is defined as:

$$\hat{\beta}^{IV} = \frac{\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0]}{\mathbb{E}[D_i | Z_i = 1] - \mathbb{E}[D_i | Z_i = 0]}.$$

Assume the following four conditions hold:

- (i) **Independence:** $(Y_i(0), Y_i(1), D_i(0), D_i(1)) \perp Z_i$
 - (ii) **Exclusion restriction:** Z_i affects Y_i only through D_i
 - (iii) **Relevance:** $\mathbb{E}[D_i | Z_i = 1] \neq \mathbb{E}[D_i | Z_i = 0]$
 - (iv) **Monotonicity:** $D_i(1) \geq D_i(0)$ for all i
- (a) (5 points) Under the assumptions above, prove that:

$$\hat{\beta}^{IV} = \mathbb{E}[Y_i(1) - Y_i(0) | D_i(1) > D_i(0)].$$

Carefully indicate the assumption that allows you to write the equation differently at each stage of the proof.

- (b) (6 points) Discuss the external validity of $\hat{\beta}^{IV}$. For which subpopulation does LATE recover the treatment effect, and under what conditions does it coincide with the Average Treatment Effect? What are the implications for policy evaluation?